

Mid Semester Examination 2025

Name of the Program: B.Tech.

Semester: VIIIth

Name of the Course: Computer Architecture

Course Code: TOE-810

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all** the questions by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1, CO2
(a)	Draw the block diagram of a digital computer & explain its various functional units.	
OR		
(b)	How to perform the following logic operations – Selective-set, selective complement, masking, clear	
	(10 marks)	CO2
(a)	Explain 16 bit instruction code for memory reference instructions with two examples. Also mention an example for each.	
OR		
(b)	Draw & explain the flowchart for instruction cycle	
Q3	(10 marks)	CO1, CO2
(a)	Explain and classify the following instruction on the basis of memory reference, register reference & Input Output reference instruction:- (i) LDA (ii) STA (iii) CMA (iv) INP (v) AND	
OR		
(b)	Perform following arithmetic subtraction using 8 bits in 2's complement form (i) 37 – 25 (ii) 25 – 37	
Q4	(10 marks)	CO2
(a)	Explain subroutine concept with the help of a block diagram.	
OR		
(b)	Explain direct, indirect and immediate addressing modes with the help of an example.	
Q5	(10 marks)	CO3
(a)	Write a program to swap two 8 bit numbers stored at given memory location.	
OR		
(b)	Write short notes on fixed point representation and floating point representation	

